Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[3]:
```
import tensorflow as tf
from tensorflow import keras
import matplotlib.pyplot as plt
import numpy as np
```

[4]:
```
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 6s 1us/step
```

[5]:
```
print("Training images:", x_train.shape)
print("Training labels:", y_train.shape)
print("Testing images:", x_test.shape)
print("Testing labels:", y_test.shape)
```

```
Training images: (60000, 28, 28)
Training labels: (60000,)
Testing images: (10000, 28, 28)
Testing labels: (10000,)
```

[6]:
```
x_train = np.asarray(x_train, dtype=np.float32)
x_train /= 255.0
x_test = np.asarray(x_test, dtype=np.float32)
x_test /= 255.0
```

[7]:
```
model = keras.Sequential([keras.layers.Flatten(input_shape=(28, 28)),
 keras.layers.Dense(128, activation='relu'),
keras.layers.Dense(10, activation='softmax') ])
```

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[9]:
```
model.compile( optimizer=keras.optimizers.SGD(learning_rate=0.01),
 loss='sparse_categorical_crossentropy', metrics=['accuracy'] )
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be us
ed. Please use WSL2 or the TensorFlow-DirectML plugin.
```

[10]:
```
history = model.fit( x_train, y_train,
epochs=10, batch_size=32, validation_split=0.1 )
```

```
Epoch 1/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 20s 10ms/step - accuracy: 0.8262 - loss: 0.7018 - val_accuracy: 0.9133 - val_loss: 0.3307
Epoch 2/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 20s 10ms/step - accuracy: 0.8999 - loss: 0.3571 - val_accuracy: 0.9278 - val_loss: 0.2647
Epoch 3/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 14s 8ms/step - accuracy: 0.9132 - loss: 0.3077 - val_accuracy: 0.9353 - val_loss: 0.2369
Epoch 4/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 15s 9ms/step - accuracy: 0.9222 - loss: 0.2776 - val_accuracy: 0.9387 - val_loss: 0.2173
Epoch 5/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 19s 8ms/step - accuracy: 0.9289 - loss: 0.2553 - val_accuracy: 0.9442 - val_loss: 0.2007
Epoch 6/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 17s 10ms/step - accuracy: 0.9329 - loss: 0.2370 - val_accuracy: 0.9478 - val_loss: 0.1916
Epoch 7/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 14s 8ms/step - accuracy: 0.9380 - loss: 0.2205 - val_accuracy: 0.9528 - val_loss: 0.1761
Epoch 8/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 21s 8ms/step - accuracy: 0.9419 - loss: 0.2068 - val_accuracy: 0.9557 - val_loss: 0.1658
Epoch 9/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 22s 9ms/step - accuracy: 0.9455 - loss: 0.1943 - val_accuracy: 0.9582 - val_loss: 0.1583
Epoch 10/10
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 15s 9ms/step - accuracy: 0.9488 - loss: 0.1833 - val_accuracy: 0.9597 - val_loss: 0.1521
```

[11]:
```
test_loss, test_accuracy = model.evaluate(x_test, y_test)
print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy * 100, "%")
```

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```
Test Loss: 0.17752689123153687
Test Accuracy: 94.8199987411499 %
```

[13]:
```
plt.plot(history.history['loss'])
plt.title('Training Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.show()
```

